

Exam 2 Review: Choosing the Right Geometric Tool

Tuesday, November 17, 2026

Visualization: State space determines the first calculation

state space	first calculation	global structure
S^1	$f(\theta) = 0$ and $\text{sign } f$	$f \neq 0$: $T = \left \int_0^{2\pi} \frac{d\theta}{f(\theta)} \right $
$\mathbb{R}^2, \dot{z} = Az$	$\tau = \text{tr } A, \Delta = \det A,$ $\tau^2 - 4\Delta$	eigendirections and orientation
$\mathbb{R}^2, \dot{z} = F(z)$	$F(z) = 0, DF(z_*)$	nullclines, invariant sets, energy, symmetry
$S^1 \times \mathbb{R}$	$F(z) = 0, DF(z_*)$	periodic identification and energy levels, when available

Observation: Scope of nonlinear linearization

At a hyperbolic equilibrium of a nonlinear planar system, the nonlinear flow is locally topologically equivalent to its linearization. In particular, the linearization correctly determines whether the equilibrium is a sink, a source, or a saddle. At a nonhyperbolic equilibrium, the theorem is inconclusive.

Question: A phase-locking threshold on the circle

For

$$\dot{\theta} = \mu - \sin \theta, \quad \theta \in S^1,$$

find and classify all fixed points for every $\mu \in \mathbb{R}$.

Solution. Let

$$f(\theta) = \mu - \sin \theta.$$

If $|\mu| < 1$, define

$$\alpha = \arcsin \mu, \quad -\frac{\pi}{2} < \alpha < \frac{\pi}{2}.$$

The two fixed points modulo 2π are

$$\theta_s = \alpha, \quad \theta_u = \pi - \alpha.$$

Since

$$f'(\theta) = -\cos \theta,$$

Question: A phase-locking threshold on the circle (continued)

we have

$$f'(\theta_s) < 0, \quad f'(\theta_u) > 0.$$

Thus θ_s is asymptotically stable and θ_u is unstable. If $|\mu| > 1$, there are no fixed points, and every solution rotates around the circle. At $\mu = 1$ the fixed points coalesce at $\theta = \pi/2$; at $\mu = -1$ they coalesce at $\theta = 3\pi/2$. In both cases

$$f = 0, \quad f_\theta = 0, \quad f_{\theta\theta} \neq 0, \quad f_\mu = 1 \neq 0,$$

so each threshold is a saddle-node bifurcation on S^1 . At $\mu = 1$ the fixed point attracts from the left and repels from the right; at $\mu = -1$ it repels from the left and attracts from the right. Therefore each threshold fixed point is half-stable and Lyapunov unstable.

Question: Rotation period beyond phase locking

For

$$\dot{\theta} = \mu - \sin \theta, \quad \theta \in S^1, \quad |\mu| > 1,$$

compute the positive rotation period.

Solution. If $\mu > 1$, then $\mu - \sin \theta > 0$ and

$$\begin{aligned} T &= \int_0^{2\pi} \frac{d\theta}{\mu - \sin \theta} \\ &= \int_{-\pi/2}^{3\pi/2} \frac{d\phi}{\mu - \cos \phi} \quad (\phi = \theta - \pi/2) \\ &= \int_{-\pi}^{\pi} \frac{d\phi}{\mu - \cos \phi} \quad \text{by } 2\pi\text{-periodicity} \\ &= \int_{-\infty}^{\infty} \frac{2 ds}{(\mu - 1) + (\mu + 1)s^2} \quad \left(s = \tan \frac{\phi}{2}\right) \\ &= \frac{2\pi}{\sqrt{\mu^2 - 1}}. \end{aligned}$$

If $\mu < -1$, the orientation reverses and the positive period is again

$$T = \frac{2\pi}{\sqrt{\mu^2 - 1}}.$$

Notation: Trace–determinant data

For a real 2×2 matrix A , write

$$\tau = \operatorname{tr} A, \quad \Delta = \det A, \quad D = \tau^2 - 4\Delta.$$

Here D is the discriminant of the characteristic polynomial

$$\lambda^2 - \tau\lambda + \Delta.$$

Visualization: Trace–determinant classification

For $\dot{z} = Az$,

condition	type
$\Delta < 0$	saddle
$\Delta > 0, D > 0, \tau < 0$	asymptotically stable node
$\Delta > 0, D > 0, \tau > 0$	unstable node
$\Delta > 0, D < 0, \tau < 0$	asymptotically stable spiral
$\Delta > 0, D < 0, \tau > 0$	unstable spiral
$\Delta > 0, D < 0, \tau = 0$	linear center

Recall: Degenerate linear cases

When $D = 0$ and $\Delta > 0$, eigenvectors distinguish a star from a degenerate node. When $\Delta = 0$, the kernel of A is nontrivial, so the linear system has a non-isolated set of equilibria.

Question: Recover a complete linear portrait

Classify and orient

$$\dot{z} = Az, \quad A = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}.$$

Solution. We compute

$$\tau = 2, \quad \Delta = 5, \quad D = 2^2 - 4(5) = -16.$$

The eigenvalues are

$$\lambda = 1 \pm 2i,$$

so the origin is an unstable spiral. At $(1, 0)$,

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

which points upward; the rotation is counterclockwise. Since

$$A = I + 2 \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

the solution operator is

$$e^{At} = e^t \begin{pmatrix} \cos 2t & -\sin 2t \\ \sin 2t & \cos 2t \end{pmatrix}.$$

Thus radii grow like e^t and the angular velocity is 2.

Lemma: An invariant zero set is a uniqueness barrier

Let $G, H : \mathbb{R}^2 \rightarrow \mathbb{R}$ be locally Lipschitz, and consider the system

$$\dot{x} = xG(x, y), \quad \dot{y} = H(x, y).$$

Then the line $x = 0$ is invariant. No trajectory can cross from $x > 0$ to $x < 0$.

Proof. Let $y(t)$ solve

$$\dot{y} = H(0, y), \quad y(t_0) = y_0.$$

Then $(0, y(t))$ solves the full system through $(0, y_0)$. Local uniqueness forces every solution through that state to remain on the axis. A trajectory crossing from $x > 0$ to $x < 0$ would meet the axis and then leave it, which is impossible. $\square$

Question: When does an imaginary linear pair give a nonlinear center?

Analyze

$$\dot{x} = y, \quad \dot{y} = -x - x^3.$$

Determine the global phase curves and classify the origin.

Solution. The origin is the only equilibrium. Its Jacobian is

$$DF(0, 0) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

with eigenvalues $\pm i$, so linearization alone is inconclusive. Define

$$E(x, y) = \frac{1}{2}y^2 + \frac{1}{2}x^2 + \frac{1}{4}x^4.$$

Then

$$\begin{aligned} \dot{E} &= y\dot{y} + (x + x^3)\dot{x} \\ &= y(-x - x^3) + (x + x^3)y \\ &= 0. \end{aligned}$$

For every $h > 0$, there is a unique number $A_h > 0$ satisfying

$$\frac{1}{2}A_h^2 + \frac{1}{4}A_h^4 = h.$$

The level $E = h$ consists of the two graph branches

$$y = \pm \sqrt{2h - x^2 - \frac{1}{2}x^4}, \quad |x| \leq A_h,$$

which meet only at $(\pm A_h, 0)$. Hence the level is a compact simple closed curve. Moreover,

$$\nabla E = (x + x^3, y)$$

vanishes only at the origin, which does not lie on a positive level. Therefore every level $E = h$, $h > 0$, is a periodic orbit, and the origin is a nonlinear center.

Question: Period of the nonlinear oscillator

For

$$\dot{x} = y, \quad \dot{y} = -x - x^3,$$

set up the period of the orbit through $(A, 0)$, where $A > 0$.

Solution. For

$$E(x, y) = \frac{1}{2}y^2 + \frac{1}{2}x^2 + \frac{1}{4}x^4,$$

the orbit through $(A, 0)$ has energy

$$E_A = \frac{1}{2}A^2 + \frac{1}{4}A^4.$$

On the branch with $y > 0$,

$$y = \sqrt{2E_A - x^2 - \frac{1}{2}x^4}, \quad dt = \frac{dx}{y}.$$

Fourfold symmetry gives

$$T(A) = 4 \int_0^A \frac{dx}{\sqrt{2E_A - x^2 - \frac{1}{2}x^4}}.$$

With $x = A \sin \phi$,

$$2E_A - x^2 - \frac{1}{2}x^4 = A^2 \cos^2 \phi \left(1 + \frac{A^2}{2} + \frac{A^2}{2} \sin^2 \phi \right),$$

so

$$T(A) = 4 \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 + \frac{A^2}{2} + \frac{A^2}{2} \sin^2 \phi}}.$$

In particular,

$$\lim_{A \rightarrow 0} T(A) = 2\pi.$$

Question: What changes when linear damping is added?

For

$$\dot{x} = y, \quad \dot{y} = -x - x^3 - \varepsilon y, \quad \varepsilon > 0.$$

Compute $\dot{E}$ and determine whether the damped system can have a nonconstant periodic orbit.

Solution. For

$$E(x, y) = \frac{1}{2}y^2 + \frac{1}{2}x^2 + \frac{1}{4}x^4,$$

we obtain

$$\dot{E} = y(-x - x^3 - \varepsilon y) + (x + x^3)y = -\varepsilon y^2 \leq 0.$$

If a nonconstant solution were periodic with period T , then

$$0 = E(x(T), y(T)) - E(x(0), y(0)) = -\varepsilon \int_0^T y(t)^2 dt,$$

so $y \equiv 0$. The equation

$$\dot{y} = -x - x^3 = -x(1 + x^2)$$

Question: What changes when linear damping is added? (continued)

then forces $x \equiv 0$. Therefore the damped system has no nonconstant periodic orbit.

Recall: Regular level set

Let $U \subseteq \mathbb{R}^2$ be open, let $E \in C^1(U)$, and let $c \in \mathbb{R}$. The level set $E = c$ is **regular** if

$$\nabla E(z) \neq 0 \quad \text{at every } z \in U \text{ satisfying } E(z) = c.$$

Visualization: Hypotheses and their valid conclusions

Let $U \subseteq \mathbb{R}^2$ be open and let $F : U \rightarrow \mathbb{R}^2$ be a vector field.

calculation	valid conclusion
z_* is an equilibrium, $DF(z_*)$ has eigenvalues $\pm i$ $H \in C^1(U)$, $H^{-1}(0)$ regular, $\nabla H \cdot F = 0$ on $H^{-1}(0)$, F locally Lipschitz $E \in C^1(U)$, $\dot{E} = 0$ $E \in C^1(U)$, $\dot{E} < 0$ away from equilibria F locally Lipschitz	linearization is inconclusive $H^{-1}(0)$ is invariant each trajectory stays in one connected component of an energy level no nonconstant periodic orbit solution curves that meet agree after a time shift

Visualization: Invalid converses and overclaims

Let F be the vector field, and let H and E be differentiable real-valued functions on its phase space.

$$\begin{aligned}
 \lambda = \pm i &\not\Rightarrow \text{nonlinear center,} \\
 H^{-1}(0) \text{ invariant} &\not\Rightarrow H^{-1}(0) \text{ is the global basin boundary,} \\
 \dot{E} = 0 &\not\Rightarrow \text{every connected component of an energy level is a periodic orbit,} \\
 \text{numerical portrait} &\not\Rightarrow \text{proof.}
 \end{aligned}$$